

Lab_DFT_1

You have to synthesize and insert DFT logic using **DFT Compiler shell** and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- **scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db**
- **scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c**
- **scmetro_tsmc_cl013g_rvt_ss_1p08v_125c**

Steps: -

1. Copy **Lab_DFT_1** folder from your PC into the virtual machine inside **Labs** directory
2. Perform RTL preparation
 - Add scan ports (SI, SE, SO, test_mode, scan_clk, scan_rst)
 - Instantiate mux2X1 module for clocks
 - Instantiate mux2X1 module for resets
 - Connect muxed clk and muxed rst to the Instantiated modules
3. Open **dft_script.tcl** and **uncomment** reading the verilog module "mux2X1".
4. Run **dft_script.tcl** using (**dc_shell -f dft_script.tcl | tee dft.log**) to check that the instantiated file is consistent with the RTL files (**No Errors**)
5. Open **cons.tcl** and add the following constraints: -
 - Create a clock on scan_clk port with frequency 1 MHz
 - Create **clock uncertainty for setup** with 0.2 ns for scan_clk
 - Create **clock uncertainty for hold** with 0.1 ns for scan_clk
 - Create a **clock transition** with 0.05 ns for scan_clk
 - Add **set_dont_touch_network** on scan_clk
 - Define the relationship between functional and scan clock as asynchronous clocks using **set_clock_groups** command
 - Add **Input delays** on scan input ports with 20% of dft clock period
 - Add **output delays** on scan output ports with 20% of dft clock period
 - Add **Buffer driving cell** for all scan input ports
 - Add **load** of 0.5 pf on all scan output ports
 - Add **set_case_analysis** command on test_mode port with value equals **0** to run timing analysis using the functional clock (fast) after dft insertion.

6. Open dft_script.tcl and Add the following commands: -

- Configuring scan chain style using **set_scan_configuration** command
- Test-Ready Compile using **compile -scan** command
- Defining the DFT Signals (SI, SE, SO, test_mode, scan_clk, scan_rst) using **set_dft_signal** command
- Create Test Protocol using **create_test_protocol** command
- Perform Pre-DFT Design Rule Checking using **dft_drc -verbose** command
- Preview DFT using **preview_dft -show scan_summary** command
- Insert DFT logic using **insert_dft** command
- Optimize Post DFT Insertion using **compile -scan -incremental** command
- Perform Post-DFT Design Rule Checking using **dft_drc -verbose -coverage_estimate** command

7. Run dft_script.tcl using (**dc_shell -f dft_script.tcl | tee dft.log**) and check the log (dft.log) to be sure that: -

- **No Errors**
- **No Warnings**
- **No Latches**
- **No Comb Loops**

8. Check DFT Test Coverage.

9. Check setup & hold timing reports for any violations as well as Area report to find out the impact of scan flops on total design area.

10. Open Design Compiler GUI, View the DFT Scan chain.